



**POLITECNICO
MILANO 1863**

**MULTIMODAL INTERACTIVE
TECHNOLOGIES AND
APPLICATIONS**

Main Tasks for Heuristic Evaluation

Botanical Garden

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This document describes the three interaction tasks for heuristic evaluation, specifying the user goal, modalities, expected flow and success criteria for each.

1 Task 1 — Disambiguation of Overlapping Plants

Goal. Identify **one specific plant** from a cluster of visually overlapping specimens (e.g. Lavandula, Nephrolepis and Santolina in scene 1).

Modalities. *Input:* raycasting + speech (STT) + overlay click fallback. *Output:* 3D highlight on each candidate, A/B labelled overlay, TTS clarification prompt.

Flow.

1. User clicks or speaks while gazing at the overlapping plants.
2. Frontend detects multiple hits, sorts by depth and labels candidates A, B...; backend generates a spoken clarification prompt.
3. Overlay and highlights are displayed; TTS plays the prompt.
4. User resolves by voice (“A”, plant name) or by clicking; backend matches and returns the information card.

Success criteria. Highlight and overlay appear within ~1 s; voice resolution works for natural phrasings; two retries offered before click fallback; no card shown until ambiguity is resolved.

Heuristics. Visibility of system status; match with the real world; error prevention and recovery (retry + click fallback).

2 Task 2 — Comparing the Current Plant with a Visited One

Goal. Compare a newly selected plant with one seen earlier (“*How does this compare to the lavender I just looked at?*”).

Modalities. *Input:* raycasting to select the new plant + speech (comparison query). *Output:* side-by-side comparison card and TTS summary.

Flow.

1. User has visited at least one plant; its identity is stored in the client-side visited plant list.
2. User selects a second plant and asks a comparison question (“*compare it with the lavender*”, “*which is more drought-tolerant?*”).
3. Frontend parses the intent and sends it to the comparison handler with the selected plant and visited history as context.
4. Backend resolves the target plant (by name, alias or recency), fetches both records and returns a spoken comparison summary.
5. Frontend renders a side-by-side comparison card and plays TTS.

Success criteria. Implicit references (“*the previous one*”, “*lavender*”) are resolved via the visited list; card aligns matching attributes; graceful error when no second plant can be identified.

Heuristics. Recognition rather than recall; consistency and standards; aesthetic and minimalist design.

3 Task 3 — Dynamic Information Card Driven by Spoken Intent

Goal. Obtain **targeted information** about a single plant — botanical description, medicinal use, toxicity, drought tolerance, feng-shui symbolism — without navigating menus.

Modalities. *Input:* raycasting to select the plant + speech (STT); follow-up turns keep the plant in context. *Output:* context-sensitive card (medicinal, botanical, or general), 3D highlight and TTS answer.

Flow.

1. User selects a plant; intent and topic of interest are parsed.
2. Topic is mapped to a card type (medicinal, botanical, general).
3. If the topic is unavailable, backend returns a fallback with available topics and a spoken explanation.
4. Frontend renders the information card or a fallback card whose buttons re-issue requests for the same plant.
5. Subsequent turns are contextual follow-ups using dialogue history.

Success criteria. Card type matches spoken intent; fallback never dead-ends; follow-ups stay locked to the selected plant; TTS, card and highlight stay synchronised.

Heuristics. Flexibility and efficiency; error recovery via fallback topics; consistency between voice and visual output.

Out-of-Scope

Not evaluated: scene navigation via portals; STT/TTS audio quality; LLM answer factuality.